AS · Unit 2 · WPH12 Waves & Electricity

Progressive Waves

Drop a stone into a pond and the ripples spread out, but the water itself only bobs up and down. That is the whole idea of a wave: **energy travels, the medium stays where it is**. On this page we describe waves with a handful of quantities, link them with $v = f\lambda$, learn to read wave graphs, measure the speed of sound, see why waves get weaker as they spread, and use echoes to find things we can't see.

SPEC 33 Amplitude, frequency, period, speed and wavelength

WHAT THIS SPEC POINT ASKS

Know what amplitude, frequency, period, wave speed and wavelength mean.

What a progressive wave is

- A **progressive (travelling) wave** transfers **energy** from one place to another **without transferring matter**. The particles of the medium (or the field values, for light) just **oscillate about a fixed equilibrium position**.
- Mechanical waves (sound, waves on a string) need a medium. Electromagnetic waves don't: they cross a vacuum.

The five terms

Term	Symbol, unit	Meaning
Displacement	x or s, m	distance and direction of a point from its equilibrium position
Amplitude	A, m	the **maximum displacement** from equilibrium (not peak to trough)
Wavelength	λ, m	the distance between two **adjacent points in phase**, e.g. crest to crest
Period	T, s	the time for **one complete oscillation** of a point on the wave
Frequency	f, Hz	the **number of oscillations per second** (1 Hz = 1 s^{-1})
Wave speed	v, m s^{-1}	the distance the wave (the energy) travels per second

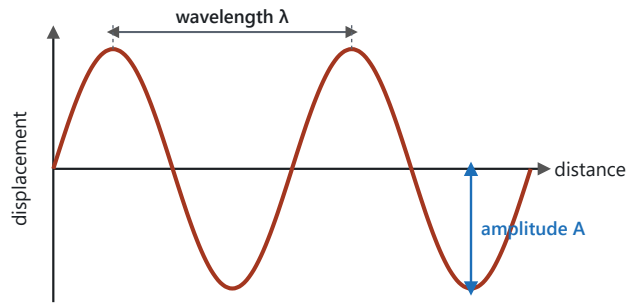
Frequency and period are reciprocals:

Period and frequency

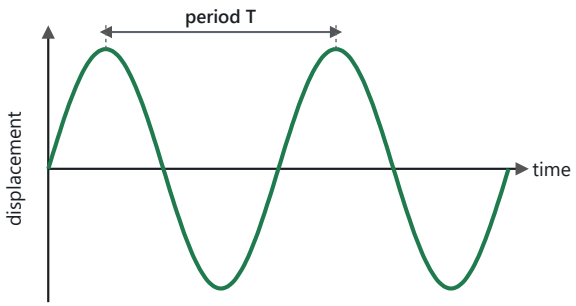
$$f = \frac{1}{T}$$

f: frequency / Hz T: period / s

Which graph gives which quantity



Displacement–distance: a snapshot of the whole wave



Displacement–time: one point on the wave

Same shape, different x-axis. Read λ off a distance axis and T off a time axis. Both show the amplitude.

Graph	Shows	Read off it
Displacement–distance	a snapshot of the whole wave at one instant	**wavelength** λ and amplitude A
Displacement–time	how **one point** moves as time passes	**period** T (so f = 1/T) and amplitude A

To find the speed you need λ from the first graph and f from the second. You can never get λ from a displacement–time graph.

Reading an oscilloscope

- Period = (number of horizontal divisions for one cycle) × (time-base setting).
- Measure across **several complete cycles** and divide. This cuts the percentage uncertainty.

WORKED EXAMPLE

frequency from an oscilloscope trace

Three complete cycles span 7.5 divisions. The time base is 2.0 ms div^{-1}.

Time for 3 cycles = 7.5 × 2.0 ms = 15 ms, so T = 15 ms ÷ 3 = 5.0 ms = 5.0×10^{-3} s.

f = 1/T = 1 ÷ (5.0×10^{-3} s) = **200 Hz** (2 s.f.).

MARK-SCHEME WORDING

- "Use of f = 1/T"; "Use of v = fλ".
- Oscilloscope questions: "uses at least 3 waves to determine T" is a marking point on its own (WPH12 Jun 2024).
- Amplitude: "maximum displacement" from the equilibrium/rest position.

Often asked as: MCQs on which graph gives which quantity, reading T from a trace, comparing two traces on one screen.

COMMON MISTAKE

Reading a "wavelength" off a displacement–time graph. The distance between peaks there is the **period**.

Measuring amplitude peak to trough (that is 2A).

Leaving the time base in ms (or μs) when working out f. Convert to seconds first.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Two sound traces on one screen, same time base: if wave 1 shows one-third as many cycles, its frequency is a third, so its wavelength is three times as long, because the speed of sound is the same for both and v = fλ (WPH12 Jun 2024).

SPEC 34 The wave equation v = fλ

WHAT THIS SPEC POINT ASKS

Use v = fλ to link wave speed, frequency and wavelength.

Where it comes from

In one period T the source makes one complete oscillation, and the wave moves forward one wavelength. Speed = distance ÷ time:

The wave equation

FORMULA SHEET

$$v = f\lambda$$

v: wave speed / m s^{-1} f: frequency / Hz λ: wavelength / m

v = wave speed (m s^{-1}), f = frequency (Hz), λ = wavelength (m). It is on the formulae sheet.

What sets what

- The **source** sets the frequency. The **medium** sets the speed. The wavelength then follows: λ = v/f.
- At a fixed speed, f and λ are inversely proportional: double f and λ halves.
- When a wave crosses into a new medium its **frequency stays the same**. Its speed changes, so its wavelength changes in proportion (light slows in glass, so λ gets shorter).

Wave	Typical speed
Sound in air (room temperature)	about 330–340 m s^{-1}
Sound in water	about 1500 m s^{-1}
Sound in steel	about 5000–6000 m s^{-1}
Any EM wave in a vacuum (and almost in air)	c = 3.00 × 10^{8} m s^{-1}

WORKED EXAMPLE

v = fλ, and a change of medium

(a) A 512 Hz tuning fork in air, v = 340 m s^{-1}: λ = v/f = 340 ÷ 512 = **0.664 m** (3 s.f.).

(b) Light of frequency 5.00 × 10^{14} Hz goes from air into water, where its speed is 2.26 × 10^{8} m s^{-1}.

In air: λ = 3.00 × 10^{8} ÷ 5.00 × 10^{14} = 6.00 × 10^{-7} m (600 nm).

In water: f is unchanged, so λ = 2.26 × 10^{8} ÷ 5.00 × 10^{14} = 4.52 × 10^{-7} m = **452 nm** (3 s.f.).

Deriving v = fλ

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Speed is distance travelled divided by the time taken:

$$v = \frac{\text{distance}}{\text{time}}$$

2. Watch one crest. In one period T the source makes exactly one complete oscillation, so it sends out exactly one more wavelength. The crest moves forward by exactly one wavelength λ in that time:

$$v = \frac{\lambda}{T}$$

After a time T, the particle one wavelength ahead is doing exactly what this one was doing, so the whole pattern has shifted along by λ.

3. Since $f = 1/T$, dividing by T is the same as multiplying by f:

$$\boxed{v = f\lambda}$$

WORKED EXAMPLE

A loudspeaker emits a note of frequency 440 Hz. The speed of sound is 340 m s^{-1} in air and 1500 m s^{-1} in water.
(a) Calculate the wavelength of the sound in air.
(b) The sound passes into a swimming pool. State its frequency in the water and calculate its wavelength there.

Show solution

(a) Rearrange $v = f\lambda$:

$$\lambda = \frac{v}{f} = \frac{340}{440} = 0.773 \text{ m } (3 \text{ s.f.})$$

(b) The frequency is set by the source and doesn't change at the boundary: $f = 440$ Hz. The speed increases, so the wavelength increases in proportion:

$$\lambda = \frac{1500}{440} = 3.41 \text{ m } (3 \text{ s.f.})$$

MARK-SCHEME WORDING

- "Use of v = fλ" is almost always the first mark, credited even if only implied by the numbers.
- In graph questions, a plot of λ against 1/f is a straight line through the origin and "gradient = v" must be stated.

Often asked as: one-step calculations with prefixes (kHz, MHz, nm), λ found from a stationary-wave pattern first, comparing travel times of light and sound.

COMMON MISTAKE

Thinking the frequency changes when a wave enters a new medium. It doesn't: v and λ change together (WPH12 Jan 2020 examiner report).

Prefix slips: 1 nm = 10^{-9} m, 1 μs = 10^{-6} s, 1 MHz = 10^{6} Hz.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Unfamiliar wave-speed formulas given in the question, which you just substitute into: deep-water waves $v = \sqrt{(g\lambda/2\pi)}$ and shallow-water waves $v = \sqrt{(gd)}$ (WPH12 Jan 2020).

Seismic S-waves have $v = \sqrt{(G/\rho)}$. A liquid has G = 0, so S-waves can't travel through liquids (WPH12 Jun 2022).

SPEC 35 Longitudinal waves

WHAT THIS SPEC POINT ASKS

Describe a longitudinal wave using the displacement of the molecules and the variation in pressure.

What it is

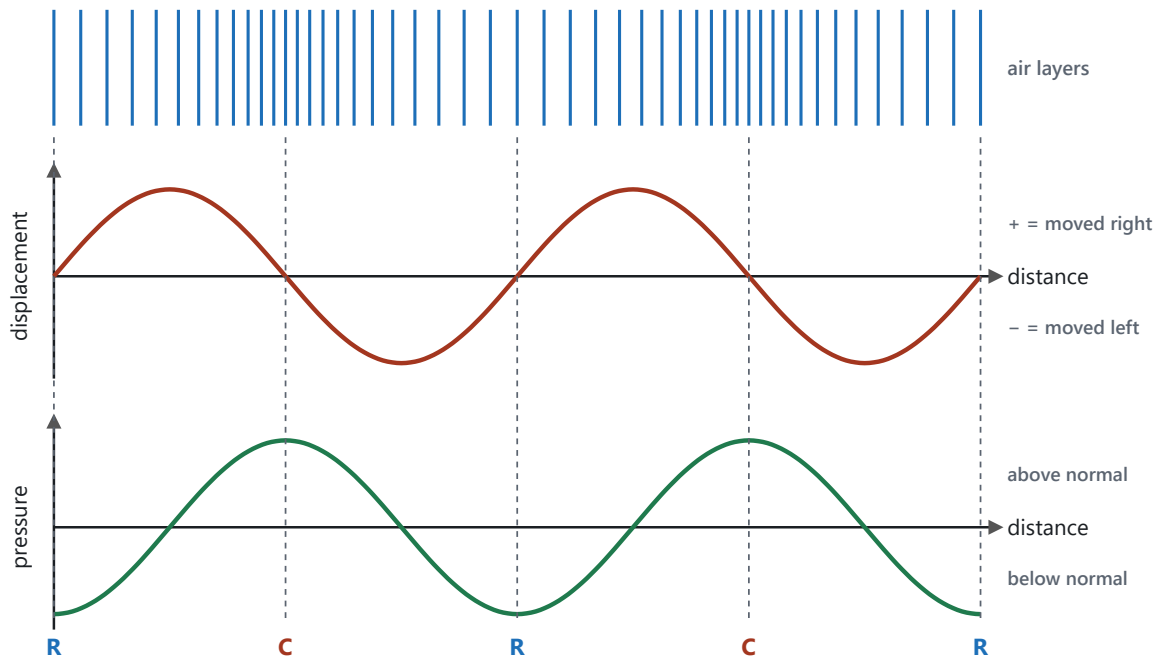
- In a **longitudinal** wave the particles oscillate **parallel to the direction the wave travels** (the direction of energy transfer).
- Examples: **sound**, ultrasound, seismic P-waves, a slinky pushed back and forth along its length.
- They need a medium, and they **cannot be polarised**.

Compressions and rarefactions

- A **compression** is a region where the molecules are pushed **closer together**, so the **pressure is above normal** (a pressure maximum).
- A **rarefaction** is a region where the molecules are **further apart**, so the **pressure is below normal** (a pressure minimum).
- Each layer of air moves forwards and backwards about its own rest position. No air travels with the wave.
- One wavelength = the distance from one compression to the next (or one rarefaction to the next).

Displacement and pressure are a quarter-cycle apart

Look at one compression. The air on its left has moved **right** (towards it) and the air on its right has moved **left** (towards it). The layer right at the centre has been squeezed from both sides but **has not moved**: its displacement is **zero**. The same is true at the centre of a rarefaction (neighbours have moved away from it). Maximum displacement is halfway between a compression and a rarefaction, where the pressure is normal.



C = compression (pressure maximum), R = rarefaction (pressure minimum).
Displacement is zero at the centre of every C and every R.

The layers bunch at C and spread at R. Displacement is zero at every C and every R; pressure is greatest at C and least at R. The two graphs are a quarter of a wavelength apart.

WORKED EXAMPLE

A sound wave has frequency 500 Hz. On its displacement–distance graph, the centre of a compression and the centre of the nearest rarefaction are 0.34 m apart. Calculate the speed of the sound.

Show solution

A compression and the next rarefaction are half a wavelength apart (R to T in the figure), so

$$\lambda = 2 \times 0.34 = 0.68 \text{ m}$$

$$v = f\lambda = 500 \times 0.68 = 340 \text{ m s}^{-1}$$

MARK-SCHEME WORDING

- Longitudinal: "oscillations/vibrations are **parallel** to the direction of wave travel / energy transfer / propagation".
- "Zero displacement at **all** compressions **and** rarefactions."
- Rarefactions are "points where the **pressure** is a minimum" (not where displacement is a minimum).

Often asked as: MCQs on where displacement is zero; sketch the displacement–distance graph from a pressure–distance graph; define and compare transverse and longitudinal.

COMMON MISTAKE

Thinking only the centre of a compression has zero displacement. The centre of every rarefaction does too (WPH12 Oct 2022 examiner report).

Assuming a sine-shaped graph means the wave is transverse. A displacement–distance graph of sound is a graph, not a picture.

Writing "the particles move along with the wave". They oscillate about fixed positions.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Comparing a sound wave with a stationary wave on a string (5 marks): same frequency; different speeds and wavelengths; every point of the sound wave has the same amplitude but the stationary wave's amplitude varies; the sound wave transfers energy, the stationary wave doesn't; string transverse, sound longitudinal. Don't contradict yourself ("different speeds but the same wavelength" can't be true at the same frequency) (WPH12 Jun 2023).

SPEC 36 Transverse waves

WHAT THIS SPEC POINT ASKS

Describe transverse waves.

What it is

- In a **transverse** wave the oscillations are **perpendicular to the direction of wave travel** (the direction of energy transfer).
- Highest points are **crests (peaks)**, lowest points are **troughs**.
- Examples: **all electromagnetic waves** (light, radio, X-rays), waves on a string or rope, seismic S-waves, a slinky shaken side to side.
- Transverse waves **can be polarised** (49). That is the one behaviour that separates them from longitudinal waves.

	Transverse	Longitudinal
Oscillation direction	perpendicular to travel	parallel to travel
Features	crests and troughs	compressions and rarefactions
Examples	EM waves, string, S-waves	sound, ultrasound, P-waves
Polarisation	yes	**no**
Reflection, refraction, diffraction, superposition, interference	yes	yes

Which way does a point move next?

On a displacement–distance snapshot of a transverse wave moving to the right:

1. Redraw the wave shifted a tiny distance to the right.
2. See whether the point's new displacement is higher or lower. That is the way it moves now.
3. Points at a crest or trough are momentarily **stationary**; points at zero displacement move **fastest**.

MARK-SCHEME WORDING

- "Oscillations/vibrations **perpendicular** to the direction of wave travel / energy transfer / propagation" (1 mark).

- "Only transverse waves can be polarised." Refraction, diffraction and superposition happen for both kinds.

Often asked as: 1-mark definitions, transverse v longitudinal (2 marks), MCQs on which property needs a transverse wave.

COMMON MISTAKE

"Perpendicular to the wave" on its own is vague. Say perpendicular to the **direction of travel / energy transfer**.

Saying all transverse waves travel at the same speed. Only EM waves share a speed (in a vacuum).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Radio waves from an aerial are plane polarised, so a receiving aerial must be aligned with the transmitting one, acting like a polarising filter (WPH12 Jan 2026).

SPEC 37 Graphs of transverse, longitudinal and stationary waves

WHAT THIS SPEC POINT ASKS

Draw and interpret graphs of transverse and longitudinal waves, including stationary waves.

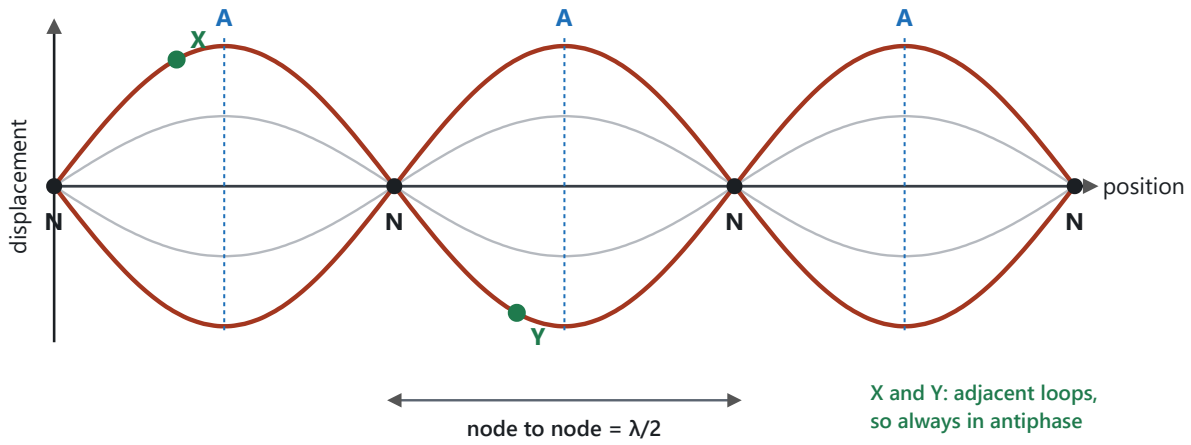
The checklist for any wave graph

1. **Read the axes first.** Distance on the x-axis → λ; time on the x-axis → T.
2. **Displacement** on the y-axis gives the amplitude. **Pressure** on the y-axis (sound only) shows compressions as maxima.
3. For a longitudinal wave, + displacement means "moved in the direction of travel", − means "moved back". The graph is not a picture of the wave.

Longitudinal: converting between the two graphs

- Pressure maxima (compressions) and minima (rarefactions) are where the displacement graph crosses **zero**.
- So the displacement graph is the pressure graph shifted by **a quarter of a wavelength** (see the figure in 35).
- When sketching: mark every C and R, put a zero of displacement at each, and draw a full sine curve through them with the same wavelength.

Stationary waves on a graph



Snapshots of a stationary wave. The red curves are the extreme positions; the grey ones are in-between instants. Nodes (N) never move. X and Y are in adjacent loops.

- The amplitude **depends on position**: zero at nodes, maximum at antinodes.
- Every point between two adjacent nodes moves **in phase** (they all reach their maximum together).
- Points in **adjacent loops** are in **antiphase** (180° apart): when one is up, the other is down.
- Twice per cycle the whole string is flat (the middle grey line), and every point is moving fastest at that moment.

WORKED EXAMPLE

reading a stationary-wave snapshot

A string shows three loops over 1.80 m at 50 Hz.

Three loops = 3 × λ/2, so λ = 2 × 1.80 ÷ 3 = 1.20 m.

Points in the first and second loops: antiphase. Two points in the same loop: in phase, but with different amplitudes unless they are the same distance from a node.

Wave speed of the travelling waves that make it: v = fλ = 50 × 1.20 = **60 m s^{-1}** (2 s.f.).

WORKED EXAMPLE

A displacement–distance graph for a wave on a rope shows crests 1.2 m apart and a maximum displacement of 3.0 cm. A displacement–time graph for one point on the rope shows the pattern repeating every 0.080 s.
(a) Calculate the frequency and the speed of the wave.
(b) How far does a crest travel in 0.20 s?

Show solution

(a) The period comes from the displacement–time graph, the wavelength from the displacement–distance graph:

$$f = \frac{1}{T} = \frac{1}{0.080} = 12.5 \text{ Hz}$$

$$v = f\lambda = 12.5 \times 1.2 = 15 \text{ m s}^{-1}$$

(b) The crest moves at the wave speed: $15 \times 0.20 = 3.0$ m. The amplitude (3.0 cm) isn't needed: it doesn't affect the speed.

MARK-SCHEME WORDING

- Phase between two points: "in antiphase" because they are "in adjacent node-to-node regions/loops" (WPH12 Jun 2022).
- Different amplitudes: "S is at an antinode **and** T is between a node and an antinode".
- Drawing: correct shape, a node at each fixed end, antinode(s) in the right place.

Often asked as: phase and amplitude of points on a stationary wave (MCQ), sketch a pattern, read an oscilloscope trace, sketch displacement from pressure.

COMMON MISTAKE

Saying points in different loops are in phase. Adjacent loops are always in antiphase (WPH12 Jan 2022 examiner report).

Saying a node has "minimum displacement". It has **zero amplitude**: its displacement is always zero.

SPEC 38 Core practical 4: speed of sound in air

WHAT THIS SPEC POINT ASKS

Find the speed of sound in air using a two-beam oscilloscope, a signal generator, a loudspeaker and a microphone.

Principle

The oscilloscope shows two traces at once: the signal driving the loudspeaker and the signal picked up by the microphone. Moving the microphone **one wavelength** further from the speaker brings its trace back to the same phase relationship with the generator's trace. Measure λ that way, read f from the generator, and use v = fλ.

CORE PRACTICAL

Core practical 4: method in brief

1. Connect the signal generator to the loudspeaker and to channel 1 of the oscilloscope. Connect the microphone to channel 2. Set f to about 2–5 kHz.
2. Put the speaker and microphone on a bench along a metre rule. Move the microphone until the two traces are **in phase** (peaks line up). Record its position.
3. Slide the microphone away from the speaker, counting each time the traces come back **in phase**. Each return = one more wavelength. Move through **several** (e.g. 5) wavelengths and record the final position.
4. λ = total distance ÷ number of wavelengths.
5. Check f by measuring T on the oscilloscope (several cycles), f = 1/T, as well as reading the generator dial.
6. Repeat for several frequencies.

Analysis: plot λ (y) against 1/f (x). It is a straight line through the origin with **gradient = v**.

Variables

- **Independent:** frequency of the signal generator.
- **Dependent:** wavelength (microphone distance moved per cycle of phase).
- **Control:** room temperature (sound speed depends on it), speaker and microphone heights, keep the same "in phase" judgement each time.

Errors and improvements

- Judging exactly when the traces line up is the biggest uncertainty. Reduce it by measuring across **many wavelengths**, and by using lower frequencies (longer λ), so the same ± distance is a smaller percentage.
- **Reflections** from the bench, walls and ceiling add extra waves that blur the pattern. Work away from walls, put soft material on the bench.
- No reaction time is involved: the oscilloscope does the timing.
- Safety: keep the volume low.

WORKED EXAMPLE

processing CP4 results

At f = 4.00 kHz the traces are in phase at 12.0 cm and again, five wavelengths later, at 54.5 cm. Each reading is ±0.5 cm.

5λ = 54.5 − 12.0 = 42.5 cm, so λ = 8.50 cm = 0.0850 m.

v = fλ = 4.00 × 10³ × 0.0850 = **340 m s⁻¹** (3 s.f.).

Uncertainty in the distance = 0.5 + 0.5 = ±1.0 cm. Over five wavelengths: 1.0 ÷ 42.5 × 100 = 2.4 %. Measured over one wavelength (8.5 cm) it would be 12 %.

WORKED EXAMPLE

In Core Practical 4 the signal generator is set to 2.50 kHz. The microphone positions at which the traces are in phase are 12.0, 25.8, 39.5, 53.4 and 67.0 cm. Find the speed of sound.

Show solution

Successive positions are one wavelength apart. Plotting position against $n = 0, 1, 2, 3, 4$ gives a best-fit gradient of 13.76 cm. Without a graph, use the first and last readings (four wavelengths apart), which uses all the spacing at once:

$$\lambda = \frac{67.0 - 12.0}{4} = 13.75 \text{ cm} = 0.1375 \text{ m}$$

$$v = f\lambda = 2500 \times 0.1375 = 344 \text{ m s}^{-1}$$

This is close to the accepted 343 m s⁻¹ at 20 °C. Averaging the four separate gaps (13.8, 13.7, 13.9, 13.6 cm) gives the same answer, because the middle readings cancel out.

MARK-SCHEME WORDING

- Method marks: microphone and speaker both on the oscilloscope; start with traces in phase (or antiphase); move the microphone until they are next in phase (antiphase); "distance moved = λ" (or λ/2 from in phase to antiphase); f from the generator or from T on the screen; then v = fλ.
- The final mark wants the **route**, e.g. "plot λ against 1/f, gradient = v" or "multiply λ by 1/T", not just "use v = fλ".
- The oscilloscope beats the ear: "hard for a person to judge when the sound is loudest/quietest"; "the oscilloscope can measure amplitude more accurately" (WPH12 Jun 2019).
- Percentage uncertainty is larger for a shorter λ (higher f). Don't write "uncertainty" when you mean "percentage uncertainty".

Often asked as: describe the method (5 marks), interpret two traces, % uncertainty at two frequencies, the two-speaker variant.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Two-speaker variant: the microphone finds quiet points where the path difference is λ/2. Path lengths come from Pythagoras, e.g. $\sqrt{(3.00^2 + 3.34^2)} - \sqrt{(3.00^2 + 1.66^2)} = 4.49 - 3.43 = 1.06$ m = λ/2, so λ = 2.12 m and v = 160 Hz × 2.12 m = 339 m s^{-1}. Reflections from walls and ceiling make the maxima and minima less clear (WPH12 Jun 2019).

Resonance-tube versions of the experiment: see 41.

SPEC 44 Intensity of radiation, I = P/A

WHAT THIS SPEC POINT ASKS

Use I = P/A for the intensity of radiation.

Definition

Intensity is the power transferred per unit area, through an area **perpendicular** to the direction the wave travels.

Intensity of radiation

FORMULA SHEET

$$I = \frac{P}{A}$$

I: intensity / W m^{-2} P: power / W A: area / m^2

I = intensity (W m^{-2}), P = power (W), A = area (m^2). It is on the formulae sheet.

The inverse square law

A point source sends its power out equally in all directions. At distance r the power is spread over the surface of a sphere, **$A = 4\pi r^2$**:

Intensity from a point source

$$I = \frac{P}{4\pi r^2}$$

I: intensity / W m^{-2} P: power of the source / W r: distance from the source / m

- Double the distance → intensity drops to **a quarter**. Triple it → a ninth.
- This assumes no energy is absorbed on the way.
- Intensity also rises with the square of the amplitude (twice the amplitude, four times the intensity).

WORKED EXAMPLE

point source and the Sun

(a) A small speaker emits 0.50 W of sound equally in all directions. At 3.0 m:

$I = P/4\pi r^2 = 0.50 \div (4\pi \times 3.0^2) = 0.50 \div 113 =$ **4.4×10^{-3} W m^{-2}** (2 s.f.). At 6.0 m it is a quarter of this, 1.1×10^{-3} W m^{-2}.

(b) Sunlight at the Earth (1.50×10^{11} m from the Sun) has intensity 1.36×10^{3} W m^{-2}. Power of the Sun:

$P = I \times 4\pi r^2 = 1.36 \times 10^{3} \times 4\pi \times (1.50 \times 10^{11})^2 =$ **3.85×10^{26} W** (3 s.f.).

WORKED EXAMPLE

tilted solar panel

Sunlight of intensity 1.00 kW m^{-2} falls on a 1.60 m² panel. The light meets the panel at 30.0° to its normal. The panel is 18.0 % efficient.

Effective area facing the Sun = $A \cos\theta = 1.60 \times \cos 30.0° = 1.386$ m².

Power arriving = $1.00 \times 10^{3} \times 1.386 = 1386$ W.

Useful output = $0.180 \times 1386 =$ **249 W** (3 s.f.).

Intensity at a distance r from a point source

1. A point source radiates power P equally in every direction. At a distance r all of that power passes through an imaginary sphere of radius r, whose surface area is

$$A = 4\pi r^2$$

2 Apply the definition of intensity to that sphere:

$$\boxed{I = \frac{P}{4\pi r^2}}$$

3 So $I \propto 1/r^2$. Double the distance and the same power covers $2^2 = 4$ times the area, so the intensity falls to a quarter. Triple it and the intensity falls to a ninth.

This assumes nothing is absorbed on the way and the source really does radiate evenly in all directions.

WORKED EXAMPLE

A small loudspeaker emits 0.50 W of sound evenly in all directions.
(a) Calculate the intensity 3.0 m from the speaker.
(b) An eardrum has an area of $6.0 \times 10^{-5}\ \text{m}^2$. Calculate the sound power entering one ear at this distance.
(c) At what distance would the intensity fall to $1.0 \times 10^{-4}\ \text{W m}^{-2}$?

Show solution

(a) The power spreads over a sphere of radius 3.0 m:

$$I = \frac{P}{4\pi r^2} = \frac{0.50}{4\pi \times 3.0^2} = 4.42 \times 10^{-3}\ \text{W m}^{-2}$$

(b) Rearrange $I = P/A$ for the small area of the eardrum:

$$P = IA = 4.42 \times 10^{-3} \times 6.0 \times 10^{-5} = 2.7 \times 10^{-7}\ \text{W}$$

(c) Rearrange for r:

$$r = \sqrt{\frac{P}{4\pi I}} = \sqrt{\frac{0.50}{4\pi \times 1.0 \times 10^{-4}}} = \sqrt{398} = 20\ \text{m}$$

Check with ratios: the intensity has dropped by a factor of 44, so the distance has grown by $\sqrt{44} = 6.6$, and $3.0 \times 6.6 = 20\ \text{m}$.

MARK-SCHEME WORDING

- "Use of I = P/A", "use of A = 4πr²" (a dimensionally correct area: not πr² for a sphere), then compare with the quoted value and conclude.
- Intensity definition: "power per unit area" or "energy per second per metre squared".
- MCQ: power constant, so I × A is constant; doubling distance quarters I.

Often asked as: on almost every paper: stars and the Sun, routers, LEDs, solar panels with efficiency, number of photons per second from an intensity (58).

COMMON MISTAKE

Using πr² instead of 4πr² for a point source.

Not converting cm² to m² (1 cm² = 10^{-4} m²). This turned 0.48 W into 48 W for many (WPH12 Jan 2025 examiner report).

Comparing different time spans (a year's energy against one hour's) or giving a power when an energy is asked for. Use E = Pt and convert hours and years to seconds.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

The **cos θ** factor for a tilted panel and **efficiency = useful power out ÷ total power in** appear regularly, e.g. 1.1 kW m^{-2} × 2.2 m² × cos 57° = 1.32 kW, of which 20 % is used (WPH12 Jun 2026).

SPEC 55 Transmission and reflection at an interface

WHAT THIS SPEC POINT ASKS

Understand that waves can be transmitted and reflected where two media meet.

What happens at a boundary

- When a wave reaches the boundary between two media in which it travels at **different speeds** (different densities or materials), part of it is **reflected** and part is **transmitted** (refracted if it arrives at an angle). Some energy may also be **absorbed**.
- The **bigger the difference** between the two media, the **more is reflected** and the less is transmitted. Similar media → mostly transmitted.
- Reflection obeys the law of reflection: angle of incidence = angle of reflection (measured from the normal).

- As a wave travels on through a medium, absorption keeps reducing its amplitude, so echoes from deep boundaries are weak.

Examples

Boundary	What happens	Consequence
air–skin (ultrasound)	very different: almost all reflected	**gel** is used to remove the air layer so the ultrasound gets in
soft tissue–soft tissue	similar: small reflection, most transmitted	each organ boundary gives a small echo, and deeper ones can still be reached
soft tissue–bone or lung (air)	very different: most reflected	little reaches what lies behind
steel–air in a cracked rail	very different	a crack gives a strong echo

MARK-SCHEME WORDING

- "Reflects at a **boundary between different materials / media / densities**"; "the speed of ultrasound in air is different from that in steel" is accepted.
- "**Some** of the pulse is transmitted / passes through."
- "Bounces back" is ignored. Write **reflected**.

Often asked as: why part of a pulse is reflected, why deeper features give weaker or no echoes, why little ultrasound reaches the lungs (MCQ).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

If the first crack reflects most of the pulse, too little energy reaches a deeper crack for its echo to be detected (WPH12 Oct 2019).

SPEC 56 Pulse-echo techniques

WHAT THIS SPEC POINT ASKS

Understand how a pulse-echo technique locates an object, and how the wavelength and the pulse duration limit the information you can get.

How it works

1. A transmitter sends out a **short pulse** of waves (ultrasound, sonar, radar, laser light).
2. The pulse is **reflected** from each boundary or object it meets.
3. A receiver detects the echoes and the **time** between sending and receiving is measured.
4. The pulse goes there **and back**, so with the wave speed known:

Distance from a pulse-echo time

$$d = \frac{vt}{2}$$

d: distance to the reflector / m v: wave speed / m s^{-1}

t: time from sending the pulse to receiving its echo / s

The time of each echo gives the depth of each boundary; the strength of the echo says something about the boundary.

What limits the detail

Limit	Why	Consequence
Wavelength	waves diffract round objects about the size of λ or smaller, so they don't reflect a clear echo	you can't resolve detail smaller than about λ. **Higher frequency → shorter λ → finer detail**. But high frequencies are absorbed more, so they don't reach as deep
Pulse duration	a long pulse means a long echo; echoes from two close boundaries overlap	two boundaries closer than about (v × pulse duration)/2 can't be told apart. **Shorter pulses → better depth resolution**
Time between pulses	each echo must return before the next pulse is sent, or you can't tell which pulse it came from	maximum range = v × (time between pulses)/2

WORKED EXAMPLE

ultrasound scan

Ultrasound in soft tissue travels at 1540 m s^{-1}.

(a) An echo returns 65 μs after the pulse: depth = vt/2 = 1540 × 65 × 10^{-6} ÷ 2 = **0.050 m (5.0 cm)**.

(b) At 3.5 MHz, λ = v/f = 1540 ÷ 3.5 × 10^{6} = 4.4 × 10^{-4} m, so detail smaller than about **0.4 mm** can't be resolved.

(c) A 2.0 μs pulse is 1540 × 2.0 × 10^{-6} = 3.1 mm long in tissue. Boundaries closer than about half this, **1.5 mm**, give overlapping echoes.

(d) Pulses sent every 0.20 ms: the deepest echo that returns in time comes from 1540 × 0.20 × 10^{-3} ÷ 2 = **0.15 m**.

WORKED EXAMPLE

An ultrasound scanner sends pulses of frequency 3.0 MHz into soft tissue, where the speed of sound is 1540 m s^{-1}. Each pulse lasts 2.0 μs. An echo from a boundary returns 65 μs after the pulse is sent.
(a) Calculate the depth of the boundary.
(b) Calculate the wavelength and the number of complete waves in each pulse.
(c) Two boundaries are 1.0 mm apart. Explain whether this scanner can show them as separate boundaries.

Show solution

(a) The pulse goes there and back, so halve the total distance:

$$d = \frac{vt}{2} = \frac{1540 \times 65 \times 10^{-6}}{2} = 0.050 \text{ m} = 5.0 \text{ cm}$$

(b) $\lambda = \frac{v}{f} = \frac{1540}{3.0 \times 10^{6}} = 5.1 \times 10^{-4} \text{ m} = 0.51 \text{ mm}$. Number of cycles = pulse duration × frequency $= 2.0 \times 10^{-6} \times 3.0 \times 10^{6} = 6$.

(c) The pulse is $v\tau = 1540 \times 2.0 \times 10^{-6} = 3.1 \text{ mm}$ long. Echoes from boundaries 1.0 mm apart arrive $2 \times 1.0 \times 10^{-3}/1540 = 1.3 \ \mu\text{s}$ apart, which is less than the 2.0 μs pulse duration. The echoes overlap and merge, so the scanner **cannot** resolve them. It would need pulses shorter than 1.3 μs (for example fewer cycles, or a higher frequency).

MARK-SCHEME WORDING

- 6-mark structure: pulses sent; **reflected** at boundaries; **time** between sending and receiving measured; speed known; distance = speed × time; **half** the time (or half the distance).
- "Uses half time or half distance" is almost always a separate mark.
- Resolution: "higher frequency, shorter wavelength", "more detail / better resolution", "smaller objects can be detected".

Often asked as: timing calculations with the factor of 2 (often ending in a claim to check), 6-mark descriptions of the technique, why a higher frequency is used, the effect of pulse length or the time between pulses.

COMMON MISTAKE

Forgetting to halve the time.

"Bounces back" instead of "reflected" (WPH12 Jan 2020 examiner report).

Not converting ms or μs to seconds (WPH12 Jun 2024 examiner report).

Saying a longer gap between pulses "increases distance" without the reason: the echo must return before the next pulse goes out (WPH12 Jun 2024).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Laser altimetry from a satellite: height = c × t/2 (e.g. 3.20 ms gives 480 km); only photons from the laser should be counted, as other light adds false signals (WPH12 Oct 2021).

Quick checks

QUICK CHECK

In a sound wave, what is the distance between the centre of a compression and the centre of the next rarefaction?

- $\lambda/4$
- $\lambda/2$
- λ
- 2λ

Compression to compression is one wavelength, and the rarefaction sits halfway between them. On a transverse wave, crest to trough is $\lambda/2$ for the same reason.

QUICK CHECK

Sound passes from air into water, where it travels about four times faster. What happens to its frequency and wavelength?

- Frequency increases, wavelength stays the same
- Frequency stays the same, wavelength decreases
- Frequency stays the same, wavelength increases
- Both increase

The source fixes the frequency, so f can't change at the boundary. Since $v = f\lambda$, a four times bigger v means a four times longer λ.

QUICK CHECK

A transverse wave travels to the right along a rope. Point P has zero displacement, and the nearest crest is just to its left. Which way is P moving at this instant?

- Upwards
- Downwards
- To the right
- It is momentarily at rest

The crest is moving right, towards P, so P is about to be at a crest: it must be rising. Particles in a transverse wave never move along the direction of travel, and they are only at rest at a crest or trough, not at zero displacement (where they move fastest).

QUICK CHECK

The intensity 2.0 m from a point source is 36 W m⁻². What is the intensity 6.0 m from the source?

- 12 W m⁻²
- 6.0 W m⁻²
- 4.0 W m⁻²
- 1.0 W m⁻²

The distance has tripled, so the area has gone up by $3^2 = 9$: $36/9 = 4.0\ \mathrm{W\ m^{-2}}$. Dividing by 3 would treat intensity as $\propto 1/r$.

QUICK CHECK

A ship's echo sounder receives the echo from the sea bed 0.40 s after sending the pulse. Sound travels at 1500 m s⁻¹ in sea water. How deep is the sea?

- 150 m
- 300 m
- 600 m
- 3750 m

$d = vt/2 = 1500 \times 0.40/2 = 300\ \mathrm{m}$. 600 m is the total distance down and back up.

QUICK CHECK

In Core Practical 4, the signal generator is set to 4.0 kHz. Through what distance must the microphone be moved for the two traces to go from in phase to in phase again? (Speed of sound = 340 m s^{-1}.)

- 4.3 cm
- **8.5 cm**
- 17 cm
- 85 cm

The traces come back into phase each time the microphone moves one wavelength: $\lambda = v/f = 340/4000 = 0.085$ m. Half of this (4.3 cm) is where they are in antiphase.